TECHNICAL TESTING

1. Black-Box Testing

This is one of the most appropriate methods for student system projects. The tester evaluates the system based on **inputs and expected outputs without examining the source code**.

Example:

Test Case	Input	Expected Output	Result
Login	Valid username/password	Dashboard displayed	Pass
Login	Invalid password	Error message displayed	Pass
QR Scan	Valid QR	Attendance recorded	Pass
QR Scan	Invalid QR	QR rejected	Pass
Duplicate Scan	Same QR scanned twice	Duplicate prevented	Pass
Attendance	Scan outside schedule	Transaction rejected/flagged	Pass
Report	Select date	Correct attendance report generated	Pass

Best for: Functional correctness.

2. Equivalence Partitioning

Divide possible inputs into groups or **classes**, then test representative values from each group.

For example, for QR codes:

Input Class	Example	Expected Result
Valid QR	Registered QR	Accepted
Invalid QR	Unregistered QR	Rejected
Expired QR	Expired QR	Rejected
Duplicate QR	Previously used QR	Rejected/flagged
Empty input	No QR detected	Error message

This reduces the number of test cases while still covering different input conditions.

3. Boundary Value Analysis

Test values at and around the **limits** of the system.

For example, if attendance is allowed from **7:00 AM to 8:00 AM**:

Test Time	Expected Result
6:59 AM	Not allowed
7:00 AM	Allowed
7:01 AM	Allowed
7:59 AM	Allowed
8:00 AM	Allowed/depends on rule
8:01 AM	Late/not allowed

This is particularly useful if your QR Attendance System has **time-based attendance rules**.

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4. Decision Table Testing

This is highly recommended if your attendance system has several rules.

For example:

Conditions:

- Valid QR?
- Registered user?
- Within schedule?
- Already recorded?

Valid QR	Registered	Within Schedule	Already Recorded	Expected Result
Yes	Yes	Yes	No	**Attendance Accepted**
Yes	Yes	Yes	Yes	**Duplicate Rejected**
Yes	Yes	No	No	**Late/Invalid**
Yes	No	Yes	No	**Rejected**
No	Yes	Yes	No	**Rejected**

This method is excellent for demonstrating that your system correctly implements **attendance business rules**.

5. Stress Testing

Stress testing determines how the system behaves when subjected to **more users or transactions than normal**.
For example:
Simulate 100, 200, or 500 users scanning QR codes within a short period.
Measure:

- Response time
- Failed transactions
- Server errors
- Database errors
- System crashes
- Attendance-record accuracy

Example:

Users	Transactions	Successful	Failed	Avg. Response
50	50	50	0	1.2 sec
100	100	99	1	1.8 sec
200	200	196	4	3.1 sec
500	500	481	19	5.7 sec

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6. Load Testing

Load testing is slightly different from stress testing. Instead of intentionally exceeding the system's limits, you test whether it performs well under its **expected workload**.

For example:

Test the system with 50 concurrent users because approximately 50 students may scan their QR codes simultaneously.

You can measure:

Response Time + Throughput + Error Rate + Resource Utilization

7. Usability Testing

Ask representative users to perform actual tasks.

For example:

Task 1: Log in
Task 2: Register a user
Task 3: Generate QR code
Task 4: Scan QR code
Task 5: View attendance
Task 6: Generate report

Measure:

- Task completion rate
- Time to complete task
- Number of errors
- Assistance required
- User satisfaction

Example:

Task	Users Tested	Successfully Completed	Success Rate
Login	20	20	100%
QR Scanning	20	19	95%
View Attendance	20	20	100%
Generate Report	20	18	90%

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8. Compatibility Testing

Test the QR Attendance System using different:

- Computers
- Smartphones
- Browsers
- Screen sizes
- Operating systems
- Network conditions

Example:

Device/Platform	Browser	Result
Windows PC	Chrome	Pass
Windows PC	Edge	Pass
Android Phone	Chrome	Pass
iPhone	Safari	Pass
Tablet	Chrome	Pass

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9. Recovery Testing

Intentionally interrupt the system and determine whether it can recover.

Examples:

- Disconnect internet during QR scanning.
- Restart the server.
- Close the browser during a transaction.
- Temporarily stop the database.
- Reconnect the network.

Then verify:

Are attendance records preserved?

This is especially useful for evaluating **Reliability**.

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10. User Acceptance Testing (UAT)

This is the final stage where actual users determine whether the system meets their needs.

Example acceptance criteria:

Criterion	Result
QR scanning works correctly	Accepted
Attendance records are accurate	Accepted
Duplicate attendance is prevented	Accepted

Reports are accurate	Accepted
System is easy to use	Accepted
System performs within acceptable response time	Accepted

Recommended Combination for Your QR Attendance System

For a **BSIT capstone/thesis**, I would recommend using these **five methods together**:

1. Black-Box Testing
→ Tests whether system functions work correctly.

2. Decision Table Testing
→ Tests attendance rules and conditions.

3. Performance/Load Testing
→ Tests system speed and capacity.

4. Usability Testing
→ Tests ease of use with actual users.

5. ISO/IEC 25010 Evaluation
→ Provides an overall software-quality assessment.